

Disclaimer: The questions given here are indicative of the format of questions in the MCQ section of the end semester exam, and not exhaustive. The final question paper for your exam will be set by the University and will have a wide range of questions. Please use this model paper in conjunction with your other course materials to prepare well for your exams.

Jain deemed to be University		Course Title: Web APIs				
S no	Question	A	B	C	D	Correct Ans
1	What is Actuator?	A sub-project of Spring Boot	A monitoring tool for databases	A framework for frontend development	An encryption algorithm	A sub-project of Spring Boot
2	Which of the following endpoints does Actuator provide to monitor application health?	/metrics	/info	/health	/auditevents	/health
3	What does Actuator's /metrics endpoint provide?	Health status of the application	General information about the application	CPU usage and memory usage	List of loggers and their current configuration	CPU usage and memory usage
4	How can you set a custom port number in Spring Boot?	Use the @CustomPort annotation in the main class	Set the port number programmatically using the SpringApplication class	Modify the server.port property in the application.properties file	Specify the port number as a command-line argument when running the application	Modify the server.port property in the application.properties file
5	What is a random port number in Spring Boot?	A port number that is automatically assigned by the operating system	A port number chosen randomly by the developer	A reserved port number for secure connections	A port number that changes every time the application is started	A port number that changes every time the application is started
6	How can you retrieve the assigned port number when using a random port in Spring Boot?	Access the System.getenv("PORT") environment variable	Use the @RandomPort annotation in the main class	Retrieve it from the server.port property in the application.properties file	Access the local server's configuration file	Retrieve it from the server.port property in the application.properties file
7	Spring Boot promotes database independence by:	Using proprietary SQL syntax	Requiring specific database drivers	Leveraging standard SQL and ORM frameworks	Restricting the use of ORM frameworks	Leveraging standard SQL and ORM frameworks
8	How can you update records in the database using JPA?	Use the update() method in JdbcTemplate	Execute an SQL UPDATE statement using JdbcTemplate	Retrieve the entity to be updated and use the save() method in JPA	Use the delete() method in JPA	Retrieve the entity to be updated and use the save() method in JPA
9	Why is data validation important in CRUD operations?	It ensures data consistency and integrity	It improves the performance of database operations	It eliminates the need for error handling	It simplifies the code structure	It ensures data consistency and integrity

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10	What is the benefit of using MongoDB over traditional relational databases?	MongoDB provides strict schema enforcement.	MongoDB supports complex transactions.	MongoDB offers a flexible data model and schema-less design.	MongoDB guarantees ACID (Atomicity, Consistency, Isolation, Durability) properties.	MongoDB offers a flexible data model and schema-less design.
11	Which Spring annotation is used to create a MongoDB client bean?	@MongoClientBean	@MongoConfig	@Bean	@MongoClient	@Bean
12	What is the purpose of the @Value annotation in Spring Boot?	It injects values from the application.properties file into Spring beans.	It specifies the primary key of an entity class in Spring Data JPA.	It enables the use of environment variables in Spring Boot applications.	It configures the connection pool for database connections.	It injects values from the application.properties file into Spring beans.
13	What is the advantage of service independence in microservices architecture?	Faster development, deployment, and scaling	Centralized data management	Strong coupling between services	Limited scalability options	Faster development, deployment, and scaling
14	How does microservices architecture handle failures or issues in one service?	It automatically fixes the issues in all services.	It isolates failures within individual services.	It halts the entire system until the issues are resolved.	It requires manual intervention to resolve the issues.	It isolates failures within individual services.
15	Which architectural style is commonly used in microservices to enable loose coupling and real-time communication between services?	Event-driven architecture	Layered architecture	Client-server architecture	Peer-to-peer architecture	Event-driven architecture
16	Which deployment model is commonly used when deploying Spring Boot applications on an external Tomcat server?	JAR	WAR	EAR	ZIP	WAR
17	By deploying a Spring Boot application as a WAR file, compatibility is ensured with:	Apache HTTP Server	Nginx	Tomcat and other servlet containers	Docker containers	Tomcat and other servlet containers
18	Which component of the Spring Boot application is responsible for initializing the application in a servlet container?	DispatcherServlet	Filter	ServletInitializer	ApplicationListener	ServletInitializer
19	What is the purpose of payload formats in web applications?	To define the structure and content of data	To handle security aspects of web requests	To manage user authentication and authorization	To improve performance and scalability	To define the structure and content of data
20	Which payload format is more widely used in modern web applications?	XML	JSON	YAML	CSV	JSON
21	What does XML stand for?	Extensible Markup Language	Extended Markup Language	eXtended Markup Logic	Excessive Markup Language	Extensible Markup Language
22	Which annotation can you use in your Spring WebFlux controller methods to automatically invoke the appropriate HTTPMessageReader for request body handling?	@RequestMapping	@RequestBody	@ResponseBody	@GetMapping	@RequestBody

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23	What is the primary role of HTTP Codecs in Spring WebFlux?	They handle the conversion of request and response bodies to and from Java objects.	They handle the routing and mapping of incoming HTTP requests to appropriate controller methods.	They provide security mechanisms for protecting Spring WebFlux applications from common vulnerabilities.	They handle the creation and management of reactive streams in Spring WebFlux applications.	They handle the conversion of request and response bodies to and from Java objects.
24	What is Thymeleaf primarily used for?	Building standalone Java applications	Generating dynamic and interactive web pages	Creating desktop applications	Managing databases	Generating dynamic and interactive web pages
25	Which Java framework is Thymeleaf well-suited for integration with?	Hibernate	JavaFX	Spring Boot	Apache Struts	Spring Boot
26	Thymeleaf templates closely resemble which type of files?	JavaScript files	XML files	JSON files	HTML files	HTML files
27	Which configuration file can be used to adjust properties for console log output in Spring Boot?	application.properties	logback.xml	log4j.properties	console-log-config.yml	application.properties
28	When using the CommandLineRunner interface, what type of argument does the run method accept?	A single integer	A list of strings	An array of integers	An array of strings	An array of strings
29	What is the primary reason for emphasizing security in web development?	To increase the complexity of web applications	To make web applications more feature-rich	To ensure data protection and user trust	To improve website design aesthetics	To ensure data protection and user trust
30	Which security threat involves injecting malicious SQL queries to manipulate or retrieve unauthorized data from a database?	Cross-Site Scripting (XSS)	Cross-Site Request Forgery (CSRF)	SQL Injection (SQLi)	Session Hijacking	SQL Injection (SQLi)